ENERGY CONSUMPTION AND THE POTENTIAL FOR CONSERVATION IN A SPRAY-DRIED COFFEE PLANT

M. OKADA, M. A. RAO, J. E. LIMA and M. TORLONI

ABSTRACT

Consumption of direct energy sources in a plant producing spray-dried coffee amounted to 12,930 kcal/kg of instant coffee (kg IC). Thermal energy sources accounted for 95% of the energy and electricity for the rest. Spray-drying was the most energy intensive operation, requiring 5,040 kcal/kg IC. Thermal energy consumption (kcal/kg IC) for roasting, extraction, and concentration was 890, 2,030, and 1,780, respectively. Thermal energy losses in the unit operations amounted to 55% of that consumed. Heat recovery and insulation techniques can be used to conserve about 70% of the losses.

INTRODUCTION

SINCE 1973, there has been a more than fivefold increase in the price of crude oil. This has had an adverse effect on the economies of most oil importing nations, especially the developing countries. It appears that these nations will continue to export raw materials and agricultural products in order to pay for the increasing cost of oil.

Brazil, a developing nation, derives about 45% of its energy needs from crude oil. About 80% of the oil consumed is imported (Goldemberg, 1978). Coffee (green beans and instant coffee) is a major export item of Brazil. A substantial portion of coffee is exported to the U.S. For example, Brazil contributed 29.6% (370 million kg) of the green coffee imported by the U.S. in 1972 (Boecklin, 1973). A total of 22 million kg of instant coffee were exported by Brazil in 1973 (Quast and Pinto Gil, 1973/74). While the amount of instant coffee from Brazil that reached the U.S. is not known, it is of interest to note that the U.S. imported 25.6 million kg of instant coffee in 1972 (Boecklin, 1973).

Several studies have been published on energy use and conservation in food unit operations and food processing plants. The unit operations studied were: sterilization (Chhinnan et al., 1978a; Singh, 1977), concentration (Schwartzberg, 1977), and dehydration (Flink, 1977). Energy studies in food processing plants can be divided into two categories: (1) those on the overall energy use, and (2) those on individual unit operations. Studies by Rao et al. (1976) and Goel et al. (1979) are examples of the first category, while those by Chhinnan et al. (1978b) and Knopf et al. (1978) are examples of the second category. All these studies have provided useful insights on energy use and the opportunities for conservation.

Studies on coffee processing

Sivetz and Foote (1963) and Sivetz (1963) described the historical and technological aspects of coffee: from horticulture to the manufacture of instant coffee (regular and decaffeinated). These authors also discussed numerous studies on coffee processing and these will not be discussed here; only studies on energy use for coffee will be discussed.

Quast et al. (1974) conducted performance tests on five coffee roasters. While the study emphasized the loss of dry matter, caffeine, and soluble solids, limited data were presented on the energy use in the two roasters. A continuous roaster with a capacity of 3,000 kg/hr was estimated to consume 120 kcal/kg of green coffee. In contrast, a 480 kg batch raoster was estimated to consume 440 kcal/kg green coffee.

The energy consumption and the energy efficiency improvement goals for the Roasted Coffee Industry (SIC 2095) in the U.S. were discussed (Anonymous, 1977). The industry comprised 213 plants engaged in roasting coffee and in the manufacture of spray-dried and freeze-dried coffee. The study was based on the data compiled by the Census of Manufacturers in 1972. For the industry as a whole, the energy consumption was 2,950 kcal/kg. However, it was noted (Anonymous, 1977) that instant coffee requires 5–8 times more energy than roasted and ground coffee. Coffee roasters consumed 36% of the total energy, boilers 48%, and electricity accounted for 16%. It was concluded that the energy use efficiency for the industry can be improved by 8% before 1980.

This study deals with the energy consumption, losses, and the potential for conservation in a plant producing spray-dried coffee. In particular, the objective was to determine the said quantities for each of the major unit operations. The results reported are part of a comprehensive study on energy use for coffee production, processing, and packaging undertaken by Okada (1979).

EXPERIMENTAL

THE STUDY was conducted in a plant that produced only spray-dried coffee in São Paulo, Brazil. The plant had a production capacity of 1,000 kg/hr. All the data used in the analysis were collected during the years 1976 and 1977. The data items were: the consumption of green coffee beans, the quantity and the moisture content of spent grounds, the concentration of fluid streams containing soluble coffee solids, the production of spray-dried coffee, the consumption of each energy source, and the operational characteristics and number of hours each major piece of equipment was used.

Unit operations and energy sources

A preliminary evaluation of energy use indicated that consumption of thermal energy sources was much higher than electricity. Figure 1 is a schematic of the unit operations in the plant and the end use of the thermal energy sources: diesel oil, kerosene, and fuel oil #6. As shown in Figure 1, all the diesel oil was used for roasting coffee and all the kerosene was used in the spray driers. The fuel oil #6 was used to generate steam in boilers. The steam was used to heat water used to extract solubles, for concentration of the extract, and to pre-heat air in a large spray-drier.

Coffee beans were roasted in four batch roasters in lots of 480 kg. Diesel oil was the only source of thermal energy used for roasting. Hammer mills were used to grind the roasted coffee. Coffee solubles were extracted using hot water (180°C) in a six-column extractor. Coffee extract (20–23°Brix) left the extractor at 110°C and the grounds (79% moisture) at 180°C. The extract was cooled and stored in stainless steel tanks prior to concentration.

Three rotating cone evaporators ('Centritherm', De Laval, Co.)

Author Rao is with the NYS Agricultural Experiment Station, Institute of Food Science, Geneva, NY 14456. Authors Okada and Lima are with the Instituto de Tecnologia de Alimentos, Caixa Postal 139, Campinas, S.P., Brazil. Author Torloni is with the Escola Politécnica, University of São Paulo, São Paulo, Brazil.

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were employed to concentrate the extract at 55°C and 50 mm Hg. Steam was used to evaporate the water. The coffee extract was concentrated to 39–44° Brix, cooled, and stored.

The chilled concentrate was pumped to the spray-driers. One large and two small spray-driers were used for drying. The large spray-drier was 26.9 m high and both the small driers were 17.6m high. Kerosene was used to heat air to 280°C in the small driers. In the large drier, steam was used to pre-heat air from 25°C to 160°C; kerosene was used to raise the temperature of the air from 160°C to 280°C.

Material and energy balances

Steam generated using fuel oil #6 was used for extraction, concentration, and to pre-heat air for the large spray-drier. The energy used (kcal) for each of these operations was determined employing energy balances. This method is well known and its use in energy studies was discussed by Rao and Katz (1976).

The amount of water used in the extractor was determined from the material balance of moisture entering with the ground coffee, and that leaving with the extract and spent grounds. Steam consumed to heat water from 25°C to 180°C was computed assuming a heat exchanger efficiency of 90%. Steam consumed for concentration was computed assuming 556 kcal are required per kg of water evaporated (Schwartzberg, 1977).

To determine the energy consumed for spray-drying, it was necessary to determine that required to pre-heat air for the large spray-drier (LD). For this purpose, the ratio of the mass of spray-dried coffee produced in the LD to that in the small spray-driers was determined to be 2.456:1.0. In addition, the ratio of the energy contribution by steam to that by kerosene in the LD was determined at 1.125:1.0. This ratio was determined from the increase in the temperature of the air by the respective heating media. From the two ratios and assuming the energy consumption per kg of instant coffee to be constant, it was deduced that in the LD the energy contributed by steam was 60.3% of that by kerosene.

Conversion factors for energy sources

Energy consumed in the plant was expressed in kilocalories (kcal). For diesel oil, propane, kerosene, and fuel oil #6, the kilocalories were calculated from their respective calorific values (Gatts et al., 1974). The electricity generated in the State of São Paulo is from hydro sources. The conversion factor employed for electricity was 1 kwh = 860 kcal.

Thermal energy losses and conservation

Thermal energy losses from discarded hot streams and uninsulated hot surfaces, as well as the potential for conservation were determined using methods described earlier (Rao and Katz, 1976; Goel et al., 1979).

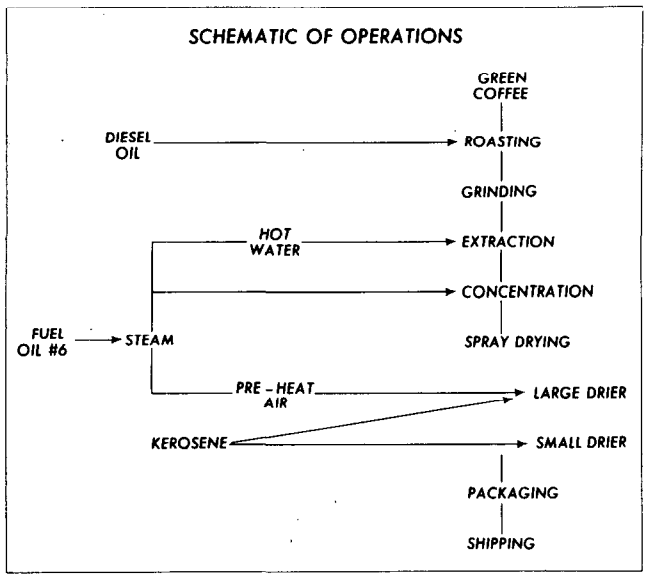


Fig. 1—Major unit operations and end use of thermal energy sources in the plant.

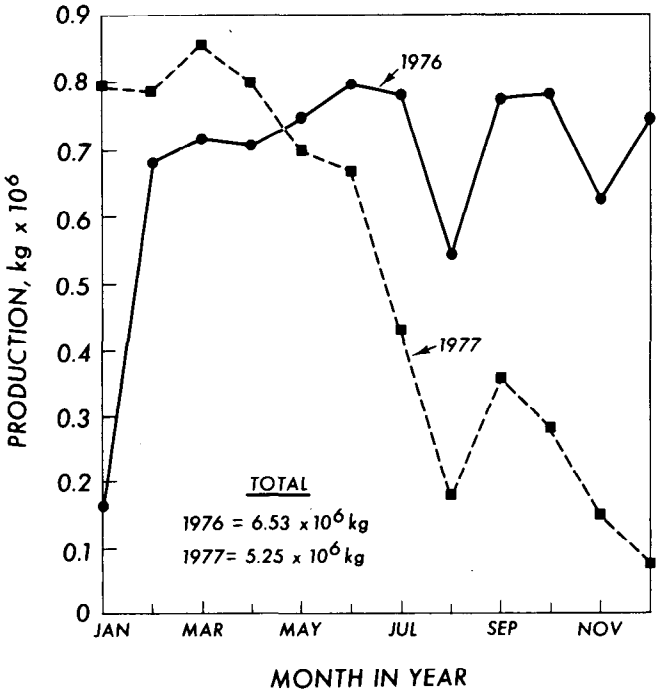


Fig. 2—Profiles of instant coffee production in the plant during 1976 and 1977.

RESULTS & DISCUSSION

Green coffee input and product output

Green coffee consumed and instant coffee produced in the plant during 1976 and 1977 are given in Table 1. The average yield of instant coffee from green coffee for the 2 yr was 37%. The spray-dried coffee produced in the plant was exported. The very fine particles trapped in the cyclone separators of the spray-driers were not included in the coffee to be exported. Unless indicated otherwise, production in the plant indicates the quantity exported. The patterns of production of instant coffee during 1976 and 1977 are shown in Figure 2. The lower production in 1977 was due to a reduction in the demand for coffee in the U.S., which in turn was the result of a coffee boycott by housewives.

Consumption of energy sources

The direct energy sources were fuels for thermal energy and electricity. Their consumption patterns during 1976 and 1977 were similar to those of production shown in Figure 2. Thermal energy was used for clean-up operations

Table 1—Consumption of green coffee and energy, and production of instant coffee in the studied plant

	1976	1977
Consumption of green coffee, kg X 10^6	17.77	14.06
Spray-dried coffee produced, kg X 10^6	6.53	5.25
Overall energy consumption,[a] kcal/kg IC		
Thermal energy sources	12,280	
Electricity	650	
Thermal energy consumption for unit operations[a], kcal/kg IC		
Coffee roasting	890	
Extraction	2,030	
Concentration	1,780	
Spray drying	5,040	

[a] Energy consumption is average for 1976 and 1977.

and in the unit operations indicated in Figure 1. Electricity was used for lighting and to drive motors.

In this discussion energy consumption is expressed in kcal per kg of instant coffee (kg IC). There was little difference in the consumption of the direct energy sources per kg IC during the 2 yr; the consumption was 5.2% higher in 1976 than in 1977. In all cases, the average for the 2 yr is given. Consumption of thermal energy and electricity per kg IC is given in Table 1.

The consumption of direct energy sources and the production during the 2 yr were found to be linearly related with a correlation coefficient, r, of 0.978. The linear equation obtained was:

$$\text{Direct energy (kcal)} = 9.28 \times 10^3 \times \text{kg IC} + 10^9$$

The intercept in the above equation can be interpreted as the energy consumed for nonproduction activities in the plant such as clean-up operations. One can expect a higher magnitude of the intercept in plants which require energy for other nonproduction activities such as space heating.

The thermal energy sources were propane, diesel oil, kerosene, and fuel oil #6. Their percentage contributions were 0.1, 10.9, 23.8, and 65.2, respectively. Electricity consumption was much lower than thermal energy (Table 1). Lighting in the plant required 29 kcal/kg IC; the rest, 621 kcal/kg IC, was used to operate electric motors.

Because of the much higher consumption of thermal energy, its consumption for the various unit operations, losses, and potential for conservation were studied. No such effort was made for electricity.

Consumption of thermal energy for unit operations

Consumption of thermal energy (kcal/kg IC) for the unit operations, roasting, extraction, concentration, and spray drying, is also given in Table 1. As expected, spray-drying was the most energy intensive operation.

The thermal energy consumption for the unit operations listed in Table 1 and for operation of fork lift trucks (propane) amounted to 9,755 kcal/kg IC. Because thermal energy consumed in the plant amounted to 12,280 kcal/kg IC, we note that 2,525 kcal need to be accounted for. Assuming the efficiency of the boilers to be 75%, one can account for about 2,000 kcal. The rest, i.e., about 500 kcal/kg IC, was used for either nonproduction activities such as clean-up or was wasted. It is emphasized that thus far energy losses associated with the unit operations were not considered. They will be discussed in a later section.

As diesel oil was the only source of thermal energy for coffee roasting, the 890 kcal/kg IC is a reliable value for coffee roasting in batch roasters. Stringent air pollution laws in the U.S. Require the use of secondary combustion equipment (afterburners) to minimize discharge of particulates. In the studied plant, the afterburners were used rarely. For this reason, it is likely the energy consumption by batch coffee roasters in the U.S. will be higher.

Energy consumed for extraction of coffee solubles was higher than for concentration and roasting. It will be shown later than a major portion of the energy is lost with the discarded coffee grounds.

As mentioned earlier, energy consumption for concentration was estimated assuming that 556 kcal are required to evaporate 1 kg of water. This was done because reliable data on concentration of coffee in rotating cone evaporators were not found in the literature.

Energy consumption for spray-drying was computed as the sum of the energy provided by steam to pre-heat air in the large spray-drier and that by kerosene. Energy provided by steam was estimated assuming that the energy required per kg IC was constant. This assumption is correct only when the concentration of the feed to the spray-drier is constant. In this study there was a small variation in the concentration of the feed. The concentration of the feed in 1976 was 32–33% solids, while that in 1977 was 33–35.7% solids.

Table 2–Thermal energy losses and potential for conservation in the major unit operations[a]

Operation	Loss[b]	Potential for conservation[c]
Roasting		
Discarded air	271 (30.3)	135 (50.0)
Surfaces of roasters	21 (2.4)	–
Extraction		
Steam condensate	487 (24.0)	438 (90.0)
Spent grounds	1,091 (53.8)	704 (64.5)
Hot surfaces	3 (0.1)	2.7 (97.0)
Concentration		
Steam condensate	308 (17.3)	276 (90.0)
Spray drying		
Steam condensate	441 (8.8)	397 (90.0)
Discarded air	1,890 (37.5)	945 (50.0)
Hot surfaces	849 (16.9)	828 (97.5)
Total for the operations	5,361 (55.0)	3,726 (69.5)

[a] Unit for energy is kcal/kg instant coffee.
[b] Numbers in parentheses are percentages of energy consumed for the operation.
[c] Numbers in parentheses are percentages of corresponding losses.

It would be desirable to determine the energy consumption for spray drying per kg of water evaporated. Such a task is difficult because the fines trapped in the cyclone separators are dissolved in water and the solution evaporated. Details of this operation are not known. Nevertheless, the water evaporated each month was estimated from the amount of coffee powder produced each month and the concentration of the feed to the spray-driers. Based on these computations, it is estimated that 2,070 kcal are required per kg of water evaporated. This estimate is in good agreement with the upper limit of 2,000 kcal/kg of water evaporated given by King et al. (1974) for dairy products.

Thermal energy losses and potential for conservation

The sources of thermal energy losses in the plant were discarded hot streams and uninsulated surfaces. Here, the sources and magnitudes of thermal energy losses for each unit operation are discussed along with the potential for conservation. The methods for conservation are heat recovery and insulation. Table 2 contains the magnitudes of the various quantities.

Roasting. The thermal energy loss in the discarded hot air of the roasters amounted to 30.3% of the energy for roasting. Loss from roaster surfaces amounted to 2.4%. The technological measures for conservation are (Anonymous, 1977): use of air pre-heaters, improving combustion efficiency, and, if possible, recycling stack exhaust gases back to the roasters. It was also suggested that scrubbers could be used to eliminate afterburners (Anonymous, 1977).

It was arbitrarily assumed that 50% of the energy in the discarded air can be recovered to pre-heat air. The duct for exhaust gases was well insulated. It was felt that additional ceramic insulation would not be justifiable.

Extraction. The condensate from the heat exchanger

used to generate hot water was discarded. The magnitude of the energy in the condensate amounted to 24% of that used for extraction. This condensate can be returned to the boilers with 90% heat recovery.

Energy lost in the discarded coffee grounds was estimated to be 53.8% of that used for extraction. One heat recovery strategy is to discharge the grounds into a chamber maintained at atmospheric pressure. By this process, 64.5% of the energy being discarded can be recovered (Okada, 1979).

The energy in the coffee extract amounted to 32.7% of that for the extraction operation. About 50% of the heat energy in the extract can be recovered by pre-heating boiler feed water. The surfaces of the extractor columns were not insulated. With 2.5 cm insulation, 96% of the losses (32.3×10^6 kcal for 1976 to 1977) can be avoided.

Concentration. The steam condensate from the concentrators was discarded; the energy in it amounted to 17.3% of that used for concentration. The condensate can be returned to the boilers recovering 90% of the energy in it.

The vapors from the evaporator were condensed by spraying water. The hot water produced was at a temperature of 40°C and a pH of about 3.7. Because of the low pH, the water cannot be used directly in the boilers. Also, stainless steel heat exchangers would be needed to recover heat from the water. For these reasons, heat recovery would be difficult.

Spray-drying. A heat exchanger was used to pre-heat air for the large spray-drier; steam was used as the heating medium. The steam condensate from the heat exchanger was discarded. Its energy content was estimated to be 441 kcal/kg IC. This condensate can also be returned to the boilers with 90% heat recovery.

Energy in the discarded exhaust air from the spray-driers was estimated to be 1,890 kcal/kg IC. Heat recovery by means of a heat exchanger would be difficult due to the presence of fine particles of coffee. The particles will clog the heat transfer surfaces. It would be preferable to scrub the exhaust air with water sprays to recover simultaneously the heat and coffee in the air stream. The heat recovery in such an operation is assumed to be 50%.

Not all sections of the spray-driers were insulated. The losses from the uninsulated surfaces for the 2 yr were estimated to be 1.0×10^{10} kcal. Providing the surfaces with 2.5 cm insulation will reduce the losses by about 98%.

CONCLUSIONS

ANALYSIS of energy consumption in a spray-dried coffee plant showed that a significant amount of energy was consumed for the manufacture of the product. Thermal energy sources contributed a major portion of the energy required. A substantial portion of the thermal energy was lost from discarded hot air and water streams, and uninsulated surfaces. Substantial energy savings can be achieved by heat recovery and insulation.

Additional savings could be achieved by using methods that require less energy. For example, a continuous roaster consumed less energy per kg green coffee than batch roasters (Quast et al. 1974). However, the loss of solids was higher in the continuous roasters. Multiple effect evaporator systems are much more efficient than the single effect evaporator used in the studied plant. However, in the manufacture of instant coffee, flavor is the overriding factor; it is more important than energy. For these reasons, coffee processors will incorporate energy conservation methods only if the quality of the product is not affected.

The energy conservation measures discussed require capital expenditures. Each conservation measure must be evaluated on the basis of the expected rate of return on the investment. Such a task was not undertaken for the studied plant. However, it must be noted that insulation and heat recovery were shown to be financially viable options in food canning plants (Rao et al., 1978).

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